EXERCISE 11.2

1. Three fair coins are tossed simultaneously. Find the probability mass function for number of heads occurred.
2. A six sided die is marked '1' on one face, '3' on two of its faces, and '5' on remaining three faces. The die is thrown twice. If X denotes the total score in two throws, find
   (i) the probability mass function (ii) the cumulative distribution function
   (iii) $P(4 \le X < 10)$ (iv) $P(X \ge 6)$
3. Find the probability mass function and cumulative distribution function of number of girl child in families with 4 children, assuming equal probabilities for boys and girls.
4. Suppose a discrete random variable can only take the values 0, 1, and 2.
   The probability mass function is defined by
   $$f(x) = \begin{cases} \dfrac{x^2+1}{k}, & \text{for } x = 0,1,2 \\ 0 & \text{otherwise} \end{cases}$$
   Find (i) the value of k (ii) cumulative distribution function (iii) $P(X \ge 1)$.
5. The cumulative distribution function of a discrete random variable is given by
   $$F(x) = \begin{cases} 0 & -\infty < x < -1 \\ 0.15 & -1 \le x < 0 \\ 0.35 & 0 \le x < 1 \\ 0.60 & 1 \le x < 2 \\ 0.85 & 2 \le x < 3 \\ 1 & 3 \le x < \infty \end{cases}$$
   Find (i) the probability mass function (ii) $P(X < 1)$ and (iii) $P(X \ge 2)$.
6. A random variable X has the following probability mass function.

x	1	2	3	4	5
$f(x)$	k^2	$2k^2$	$3k^2$	$2k$	$3k$

Find (i) the value of k (ii) $P(2 \le X < 5)$ (iii) $P(3 < X)$

7. The cumulative distribution function of a discrete random variable is given by
   $$F(x) = \begin{cases} 0 & \text{for } -\infty < x < 0 \\ \dfrac{1}{2} & \text{for } 0 \le x < 1 \\ \dfrac{3}{5} & \text{for } 1 \le x < 2 \\ \dfrac{4}{5} & \text{for } 2 \le x < 3 \\ \dfrac{9}{10} & \text{for } 3 \le x < 4 \\ 1 & \text{for } 4 \le x < \infty \end{cases}$$
   Find (i) the probability mass function (ii) $P(X < 3)$ and (iii) $P(X \ge 2)$.

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